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Optimizing Subsea Umbilical Installation Through Engineered Loop: A Practical Approach

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Abstract

Subsea umbilicals are commonly used to connect offshore facilities or subsea equipment for transferring fluid, electrical power, and communications. It is standard practice to order additional length to account for subsea conditions and installation challenges. After installation, excess length is managed through trimming, termination with the head refitted, or spooling onto a reel. Provision of an engineered loop in the routing is an alternative to managing the excess length without compromising the structural integrity, as it does not require trimming or installation of additional components.

To design an engineered loop, several factors need to be considered, i.e., shallow/deep water, seabed roughness, neighboring assets, crossings, dominant wave and current direction, and soil condition. The behavior of the engineered loop shall be studied using FE software "OrcaFlex". OrcaFlex uses structural parameters such as diameter, bending stiffness, axial stiffness, etc., along with its weight to model the actual behavior of the umbilical under the influence of the environmental loading. Dynamic analysis for a 3-hour storm shall be performed to evaluate the long-term stability of the loop.

The engineered loop profile was modeled to propose the configuration, and the results highlight an effective measure for controlling its movement and structural integrity, specifically MBR, maximum tension, and maximum displacement. Iterative simulations were conducted on multiple wave-current loading directions. Long term stability of the loop was also evaluated to determine whether secondary stabilization is required for the case in point. The results verify that the MBR, maximum tension, and maximum displacement of the umbilical in the loop area are under the acceptable criteria. The engineered loop will retain its shape with the support of secondary stabilization to control its movement. Furthermore, there is no kinking or sharp twist on the umbilical within the loop geometry. Results indicate that the loop configuration will be undisturbed, and hence the umbilical integrity is ensured. It is concluded that the umbilical engineered loop is a highly effective solution to address this installation challenge.

The proposed engineered loop solution improved on the conventional means, i.e., umbilical trimming, termination with head refitted, or spooling onto a reel, to handle the excess length. The applicability of conventional approaches is limited by the umbilical structural integrity, offshore workmanship, and lower installation productivity. Whereas an engineered loop is highly effective at addressing the installation challenge of excess umbilical length without being limited by the downsides of the conventional approaches.

Introduction

In subsea field development, the installation of umbilicals and cables often faces challenges related to excess length, seabed congestion, and route complexity. Traditionally, surplus umbilical length is accommodated through trimming, termination with the head refitted, or spooling onto a reel as a reactive solution to fit field constraints. However, the concept of an engineered loop, a deliberately designed and optimized configuration, presents a proactive approach to managing excess length while enhancing operational reliability and installation efficiency.

An engineered loop refers to a pre-planned section of umbilical layout that accommodates slack in a controlled and technically sound manner. Unlike loops that are formed during installation, engineered loops are analyzed for the dynamic stability. A well-designed loop ensures long-term functionality and structural stability.

Cable movement under environmental loading can be evaluated through the on-bottom stability methods outlined in DNV-RP-F109 ([DNV-RP-F109 2021](#)), namely generalized, absolute, and dynamic lateral stability. The generalized method, developed for pipelines, offers a straightforward and conservative approach that is useful in early design stages but often results in overdesign when applied to lighter structures such as umbilicals. The absolute lateral stability method, while adapted for umbilicals, is difficult to satisfy due to the conservative assumptions combined with the relatively low submerged weight of cables. Alternatively, the dynamic lateral stability method provides a more realistic assessment by allowing limited lateral displacement and employing advanced finite element simulations to capture the cable response under dynamic wave and current conditions ([Lauretta, Vatandoust and Thusyanthan 2025](#)).

This paper focuses on the dynamic lateral stability assessment of engineered umbilical loops using minimum bending radius (MBR), maximum tension, and maximum displacement as the primary design criteria. The dynamic analysis is conducted using OrcaFlex to evaluate the structural behavior of the loop under installation and operation conditions. Where necessary, secondary stabilizations are considered to enhance loop stability. The objective is to demonstrate that, with proper engineering and adherence to critical parameters, engineered loops can effectively accommodate excess length while maintaining long-term structural integrity and installation reliability.

A single engineered loop configuration (Omega-type) is analyzed, the objective is to assess its structural stability under different environmental load directions. Specifically, the dynamic response is evaluated under two critical wave and current headings: 90° (crossflow) and 270° (reverse crossflow). These directions are selected based on their likelihood to induce the most severe loading scenarios on the loop.

Literature Review

Managing excess length in subsea umbilicals is a practical problem that combines subsea routing, installation sequencing and component integrity. Early industry papers and field reports highlight that uncontrolled slack or large bend radii during laying/jointing create local stress concentrations, induce fatigue and risk component deformation (hydraulic tubes, electrical conductors and fibre optics) — making length management a design and operational necessity ([Offshore Magazine n.d.](#)), ([TarjomeFa n.d.](#)).

Design of the umbilical route/geometry to absorb excess length through purpose-designed subsea features (loops, spools or gentle hairpin laydowns), integrated into the subsea route to absorb excess length, are widespread in the industry ([OnePetro n.d.](#)), ([Worzyk n.d.](#)). These passive features rely on geometric design (pre-planned loops, mechanical supports) and strict enforcement of bend radius limits – to protect internal components from damage during installation or operation. Their advantage lies in simplicity and minimal intervention post installation. However, they require careful planning and design upfront to ensure safety margins are maintained throughout the design life. In contrast, umbilical management systems (UMS), present a more dynamic solution that control deployment/collection during operations. Motorised UMS units and dedicated spoolable guide/management devices either store excess umbilical or control its

footprint on the seabed during intervention and tie-in operations (Mooney 2011). These systems are in the domain of patents and niche literature and are not explored further.

From a design standpoint, recent literature highlights the importance of global umbilical seabed route optimization – coordinating umbilical loops with seabed topography, adjacent infrastructure, and hydrodynamic forces to prevent the need for re-routing or excessive bending throughout the operating lifecycle and during vessel movements (Zhang 2021). Numerical analyses and FEA-based studies reveal that decisions regarding global configuration (such as loop radius, orientation in relation to currents and nearby obstacles, and the placement of anchors/supports) significantly affect local stress distributions within umbilical components during both installation and operational phases (PRE Technologies n.d.). As a result, it is advisable to employ integrated modeling that combines large-scale routing with the response at the component level.

Practical installations described in technical proceedings demonstrate that the use of pre-designed loops or hairpin sections on the seabed can effectively manage variations in the final installed length. However, this necessitates meticulous design of loop geometry, relative orientation to dominant wave and current, and supporting geometry to prevent concentrated contact forces and to ensure minimum bend radii for enclosed hoses and tubes (Worzyk n.d.). Protective measures—such as steel arcs, engineered saddles, or seabed mattresses—are often incorporated to maintain mechanical integrity and protect against abrasion or point loads.

Recent advancements extend beyond static design. Real time monitoring techniques, including real-time spatial prediction and sensor-assisted monitoring of umbilical configuration during installation and operation (particularly beneficial when loops are utilized) enable operators to verify that loop shapes and positions remain within allowable design envelopes (Li 2024). Coupled with an integrated optimization framework for dynamic umbilicals which lends itself to more resilient loop ensuring that electrical, hydraulic, and optical performance is maintained over the asset's life (OneSubsea (SLB) n.d.).

The reviewed literature provides a clear framework for the secure and effective management of excess subsea umbilical length, integrating route-level design strategies, numerical validation, and practical installation techniques. Although existing research highlights the advantages of controlled loops, hairpin configurations, and active management systems, there is limited work focusing on the systematic incorporation of custom loop geometries into the overall mechanical design of umbilical routes, particularly under multifaceted subsea constraints. This paper seeks to fill that void by introducing a loop design methodology that is customized to specific site conditions, including bathymetry, hydrodynamic factors, surrounding assets, and interface restrictions, and is validated through comprehensive global mechanical analysis. In this manner, it enhances current industry practices by integrating excess length management into the comprehensive design process, rather than treating it as an installation modification.

Methodological Approach for Loop Design

This section outlines the approach taken to model, simulate, and evaluate engineered loops designed to accommodate excess length. The methodology is divided into the following key components:

Design Criteria

The engineered loop design process is structured to ensure that the umbilical accommodates excess length without compromising structural integrity or operational reliability. Key design parameters considered include:

- **Minimum Bending Radius (MBR):** The curvature of the umbilical must remain above the manufacturer's specified MBR to avoid damage due to excessive bending stress.
- **Maximum Tension:** Axial tension in the umbilical should remain within allowable limits to prevent tensile failure or long-term fatigue degradation.

- **Maximum Displacement:** The displacement of the loop should be controlled to avoid excessive movement that could lead to interference with nearby assets, overstressing, and instability. In accordance with DNV-RP-F109, the allowable maximum lateral displacement of the cable is defined as 10 times its outer diameter (OD).
- **Concrete Mattress as Secondary Stabilization:** The loop must remain stable during the full range of operational and installation load cases. Concrete mattresses placed over the loop are used to control vertical and lateral movement (if necessary). Their dimensions and spacing are critical to the effectiveness of the restraint system.
- **Efficient Use of Excess Length:** The loop should accommodate the required excess umbilical length with minimal footprint and complexity, providing a practical and installation-friendly alternative to conventional excess management methods.

Loop Geometry and Model Setup

The umbilical is modeled using the following mechanical properties to ensure accurate representation of the cable's structural behavior throughout the simulation:

- **Axial Stiffness:** Governs how the line resists stretching under tension.
- **Bending Stiffness:** Controls the curvature response and is crucial for evaluating MBR compliance.
- **Torsional Stiffness:** Although torsional effects are usually minimal in umbilical deployment, they are included for completeness in dynamic simulations.

These stiffness parameters are assigned uniformly along the line length, assuming a continuous and homogenous construction of the umbilical. The line weight, outer diameter, and submerged weight per unit length are also defined to reflect the actual physical characteristics of the umbilical.

The configuration features a single, rounded loop that bears a resemblance to the Greek letter (Omega). This configuration introduces moderate slack while maintaining a compact footprint and is relatively simple to implement and stabilize. The geometry is centralized and easier to confine using any additional stabilization as shown in Fig. 1 below.

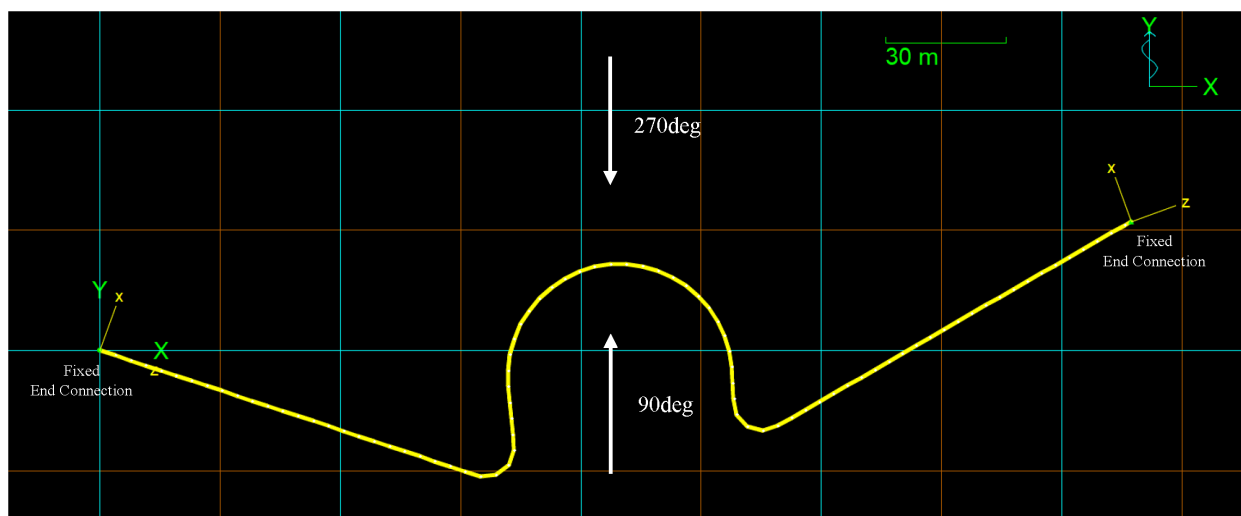


Figure 1—Omega Loop Model and Critical Direction (90deg and 270deg)

The engineered loop is positioned at the mid-span of the umbilical model. The total modeled length of the umbilical is sufficiently long to isolate the behavior of the loop. Fixed boundary conditions are applied at both ends, and adequate straight segments are included on either side of the loop to ensure that the

loop's behavior is not influenced by boundary effects. The loops are modeled using the user-specified node coordinates feature, which allows precise definition of the loop shape within the global coordinate system. This approach provides direct control over loop curvature, height, and horizontal spread.

Environmental and Load Conditions

The analysis includes operational and installation load cases based on design return periods:

- Operational conditions: 100-year RP and 10-year RP wave-current combinations
- Installation conditions: 10-year RP and 1-year RP wave-current combinations

These return period (RP) combinations are used to simulate extreme loading conditions expected during installation and operation. Each dynamic simulation spans a total of 10800 seconds (3-hour storm), with an additional 2000 seconds of ramp-up period to gradually introduce environmental loads.

Irregular sea states are often modeled using spectral formulations to capture their stochastic nature. A widely applied spectrum is the JONSWAP, which refines the classical Pierson-Moskowitz model by introducing a peak enhancement factor (γ) to represent the sharper, higher energy peaks observed in fetch-limited seas. The spectrum is given by:

$$S(\omega) = \frac{\alpha g^2}{16\pi^4} f^{-5} \exp\left\{-\frac{5}{4}\left(\frac{\omega_p}{\omega}\right)^4\right\} \gamma \exp\left[\frac{(\omega-\omega_p)}{2\sigma^2-\omega_p^2}\right]$$

Where α is the Phillips constant, g is gravitational acceleration, ω_p is the peak angular frequency, γ (typically 3.3) defines the sharpness of the spectral peak, and σ is a spectral width parameter. A random realization of the sea surface is then generated by superimposing many linear wave components with random phases, scaled according to $S(\omega)$, producing a non-repetitive time history of elevations and kinematics. This makes the JONSWAP spectrum particularly suitable for offshore engineering analyses, it provides a more realistic representation of storm-driven seas compared to simpler spectral models.

In OrcaFlex, the JONSWAP spectrum is defined by Significant wave height (H_s), Peak spectral period (T_p), and peak angular frequency (γ). These inputs allow the simulation to capture both frequency and amplitude variability in wave energy, which is essential for accurately estimating the dynamic response of flexible structures such as umbilicals. The JONSWAP parameters are specified in the environmental data section, where the software uses them to generate a time-history sea surface evaluation and corresponding wave kinematics. This enables the analysis of umbilical motion and load response under realistic and site-specific wave conditions.

The dynamic response is evaluated under two critical wave and current headings: 90° (crossflow) and 270° (reverse crossflow). These directions are selected based on their likelihood to induce the most severe loading scenarios on the loop. Refer to [Fig. 1](#) below.

Input Parameters

The following table summarizes the key mechanical properties used to model the umbilical.

Table 1—Umbilical Data

Parameter	Unit	Value
Outside Diameter	mm	89
Weight in Air (Delivery Fluids)	kg/m	10
Axial Stiffness	MN	126
Bend Stiffness	kN.m ²	5.3
Torsional Stiffness	kN.m ²	14.5
Minimum Bend Radius	m	5.5
Maximum Tension	kN	400

The environmental conditions applied in the simulation, including water depth and seawater properties, are summarized in [Table 2](#) below.

Table 2—Water Depth and Seawater Properties

Parameter	Unit	Value
Water Depth	m	50
Seawater Density	kg/m	1033
Kinematic Viscosity	m ² /s	1.51×10^{-6}

The summary of metocean conditions used for the simulation including wave and current parameters for various return periods is presented in [Table 3](#) below.

Table 3—Metocean Data Summary

Parameter	Unit	Value
1 Year RP		
Hs	m	3.1
Tp	s	7.6
Surface Current	m/s	0.7
Mid-depth Current	m/s	0.6
Bottom Current	m/s	0.4
10 Year RP		
Hs	m	4.5
Tp	s	8.7
Surface Current	m/s	0.8
Mid-depth Current	m/s	0.7
Bottom Current	m/s	0.4
100 Year RP		
Hs	m	5.7
Tp	s	9.5
Surface Current	m/s	0.8
Mid-depth Current	m/s	0.7
Bottom Current	m/s	0.4

The hydrodynamic coefficient used in this analysis is extracted from DNV-RP-C205, as shown in [Table 4](#) below.

Table 4—Hydrodynamic Coefficient

Parameter	Unit	Value
Drag Coefficient	-	1.3
Lift Coefficient	-	0.9
Added Mass Coefficient	-	2.29

Limitations

While the numerical simulation provides valuable insights into the behavior of engineered umbilical loops, several limitations should be acknowledged:

- **Simplified Seabed Representation:** The seabed is modeled as flat, without accounting for complex bathymetry.
- **No Fatigue Assessment:** The analysis does not include fatigue evaluation. Although dynamic loading is considered, long-term fatigue damage due to cyclic loads is outside the scope of this study.
- **Hydrodynamic Coefficient Assumed Constant:** Uniform values for drag and inertia coefficients were used throughout the length of the umbilical, which may not accurately reflect variation due to local geometry or flow regime.

Results and Discussion

Summary of Simulation Cases

Eight simulation cases were conducted to evaluate umbilical performance during both installation and operation conditions, under two environmental directions (90deg and 270deg). Each condition was analyzed both without additional stabilization (Scenario 1) and with the application of concrete mattresses (Scenario 2). The overview of cases is presented in [Table 5](#) below.

Table 5—Analysis Case

Case	Environmental Direction	Wave Combination	Current Combination
90deg (INS)	90deg	10-yr RP	100-yr RP
90deg (OPE)	90deg	100-yr RP	10-yr RP
270deg (INS)	270deg	10-yr RP	100-yr RP
270deg (OPE)	270deg	100-yr RP	10-yr RP

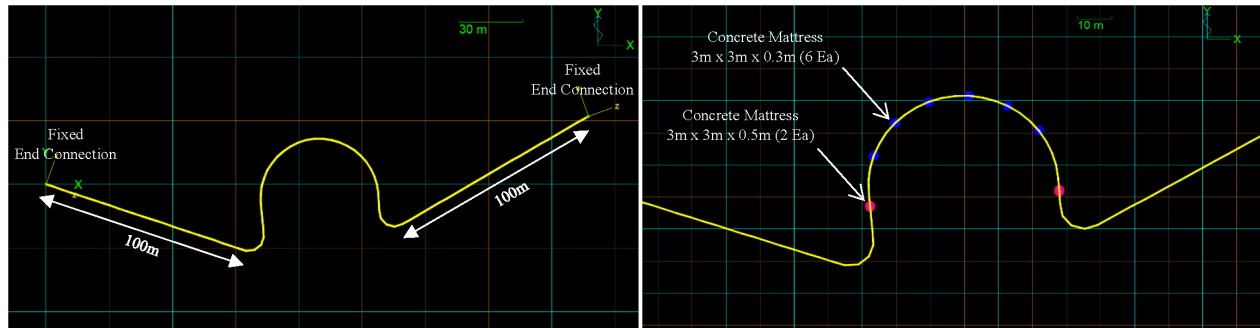


Figure 2—(a) Model Configurations (Scenario 1); (b) Model Configurations (Scenario 2)

Minimum Bending Radius (MBR)

In the no mitigation case, the calculated MBR reached 1.9m, which is substantially lower than the allowable MBR of 5m. This confirms that the configuration is unstable and does not meet the MBR requirements. In contrast, the mitigated configuration maintained the bending radius above the specified minimum, indicating adequate structural integrity. The analysis with concrete mattresses demonstrated a significant improvement in MBR values, as summarized in Table 6.

Table 6—MBR Results Summary

Case	Scenario 1 (m)	Scenario 2 (m)	Allowable (m)	Remarks	
				Scenario 1	Scenario 2
90deg (INS)	20.3	20.4	5.0	OK	OK
90deg (OPE)	9.7	11.2	5.0	OK	OK
270deg (INS)	4.5	20.3	5.0	NOT OK	OK
270deg (OPE)	1.9	20.0	5.0	NOT OK	OK

Fig. 3a (Scenario 1) and Fig. 3b (Scenario 2) illustrates the variation in MBR along the loop. The comparison highlights the effectiveness of mattress installation in maintaining the MBR above the allowable threshold.

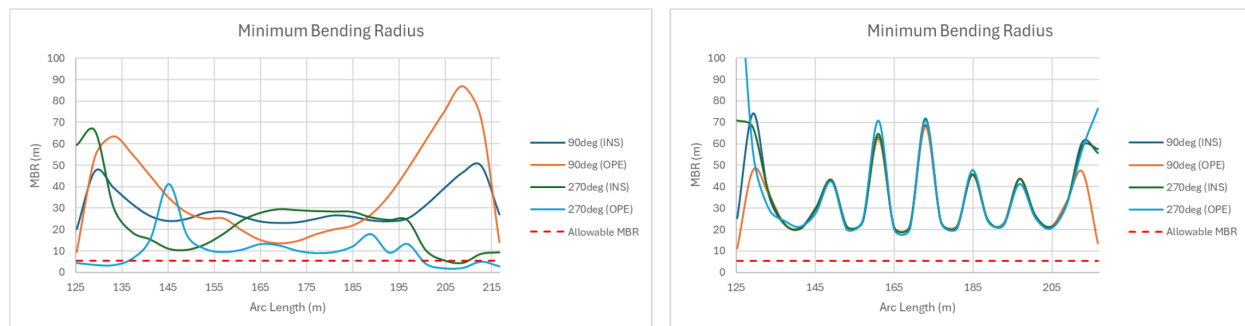


Figure 3—(a) MBR (m) Scenario 1; (b) MBR (m) Scenario 2.

Maximum Tension

The dynamic analysis of the umbilical under the selected environmental conditions shows that the maximum tension recorded is approximately 5.2kN. This value is substantially below the allowable tension limit, as summarized in Table 7 below.

This result indicates that tension is not a critical design parameter for the engineered loop in the evaluated configurations. Even under the most demanding loading directions (90deg and 270deg), the tension remains

very low relative to the structural capacity of the umbilical. This confirms that the loop configuration effectively absorbs environmental loads without introducing excessive axial forces.

Table 7—Maximum Tension Results

Case	Scenario 1 (kN)	Scenario 2 (kN)	Allowable (kN)	Remarks	
				Scenario 1	Scenario 2
90deg (INS)	2.6	1.1	400	OK	OK
90deg (OPE)	5.2	5.1	400	OK	OK
270deg (INS)	0.5	1.7	400	OK	OK
270deg (OPE)	5.0	2.6	400	OK	OK

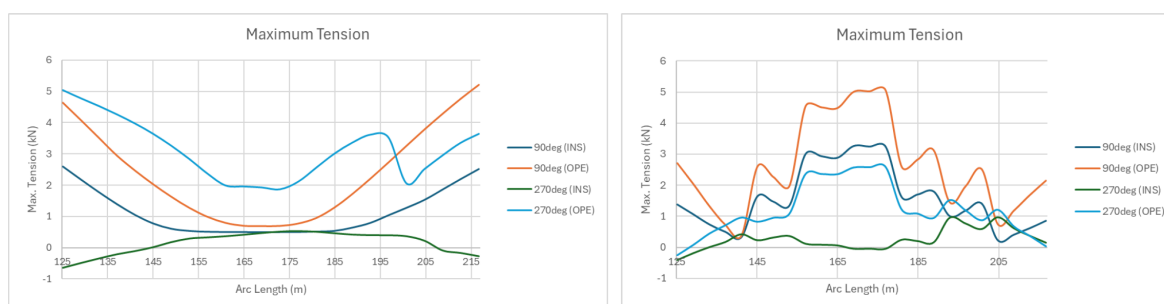


Figure 4—(a) Max. Tension (kN) Scenario 1; (b) Max. Tension (kN) Scenario 2.

Consequently, the use of engineered loops for accommodating excess umbilical length does not compromise axial load integrity and can be considered structurally conservative in terms of tension response.

Maximum Lateral Displacement

With an allowable limit of 10 OD, the no mitigation case reached 82.4 OD, far above the threshold and indicating severe instability of the loop under environmental loading. Installing concrete mattresses reduced the maximum displacement to 3.7 OD, well within the allowable range. These mattresses act as a restraint, limiting the lateral response of the loop under wave and current loading.

The result indicated in below demonstrates that with proper placement and spacing of stabilizing elements like concrete mattresses, the loop configuration remains stable and compliant with the maximum allowable displacement. This further supports the practicality of using engineered loops in dynamic subsea environments without introducing excessive movement.

Fig. 5a and Fig. 5b show the displacement distribution along the loop, highlighting the significant improvement in stability after mattress installation.

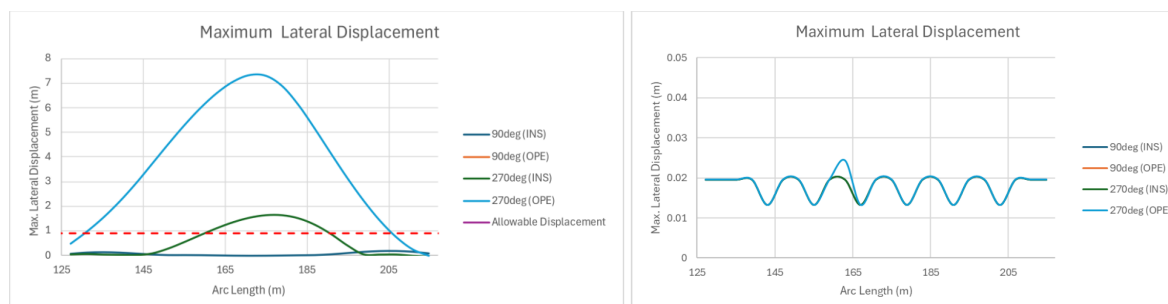


Figure 5—(a) Max. Lateral Displacement (Scenario 1); (b) Max. Lateral Displacement (Scenario 2).

Table 8—Maximum Displacement Results

Case	Scenario 1		Scenario 2		Allowable	Remarks	
	(m)	(OD)	(m)	(OD)	(OD)	Scenario 1	Scenario 2
90deg (INS)	0.2	2.0	0.3	3.1	10	OK	OK
90deg (OPE)	0.5	5.3	0.3	3.0	10	OK	OK
270deg (INS)	1.6	18.3	0.2	2.1	10	NOT OK	OK
270deg (OPE)	7.3	82.4	0.3	3.7	10	NOT OK	OK

Concrete Mattress as Secondary Stabilization

To ensure stability of the engineered loop under dynamic environmental conditions, eight concrete mattresses were placed directly over the umbilical loop section. Each mattress weighs 4.5 tons submerged, providing significant restraint to minimize vertical and lateral displacement. The simulation showed that the maximum displacement was limited to 3.7 OD, well within the allowable of 10 OD.

This highlights the importance of mattress spacing, location, and weight selection in engineered loop designs. Properly applied, this method allows for a passive and robust solution to control excess umbilical movement.

Conclusion and Recommendation

Traditional practices for accommodating excess umbilical length, such as trimming, termination with the head refitted, or spooling onto a reel. These methods can introduce higher operational risks and may affect both cost, schedules, and asset integrity particularly in complex or congested subsea developments. In contrast, the engineered loop method offers a more controlled and proactive approach to managing excess length. By absorbing the slack within a compact configuration, the integrity ensured while reducing risks associated with dynamic behavior of the umbilical.

The dynamic simulations performed in OrcaFlex confirm that engineered loops can accommodate excess umbilical length safely and reliably. With proper loop design and the use of secondary stabilization, the configuration remained stable without compromising mechanical integrity under both static and dynamic loading. Placement of mattresses directly on top of the umbilical proved effective in restraining vertical and lateral movement, therefore, reinforcing the loop's stability. There are few related works which are optional and maybe taken up on a case-by-case basis e.g., fatigue assessment under cyclic environmental loading and in-field validation using periodic ROV inspections of the engineered loop configuration.

The analysis also highlighted the importance of mattress spacing and dimensions. Inappropriate placement or sizing could result in excessive curvature or unintended displacement of the umbilical under environmental loading. When correctly applied, however, the secondary stabilization ensured the loop's integrity and confirmed the engineered loop as a practical and reliable solution for excess umbilical length management.

References

- DNV-RP-C205. 2025. Environmental conditions and loads. DNV.
- DNV-RP-F109. 2021. On-bottom stability design of submarine pipelines, cables, and umbilicals.
- Lauretta, Omar, Ali Reza Vatandoust, and Indrasenan Thusyanthan. 2025. "Insight into the on-bottom stability of cables." ISFOG 6.
- Li, W., Zhao, H., & Chen, Y. 2024. "Real-time prediction of the dynamic spatial configuration of the umbilical cable." *Journal of Marine Science and Engineering*, 12(4), 524.
- Mooney, J., Christensen, A., & Ward, M. 2011. Umbilical management system and method for subsea well ... [US Patent No. US20110198092A1]. Accessed August 08, 2025. <https://patents.google.com/patent/US20110198092A1>.
- Offshore Magazine. n.d. Deepwater installation of umbilicals - avoiding stress. Accessed August 8, 2025. <http://www.offshore-mag.com/>.

- OnePetro. n.d. OTC-30667-MS: How Technology Enables a Lower Cost for Subsea ... Offshore Technology Conference. Accessed August 08, 2025. <https://onepetro.org/OTCONF/>.
- OneSubsea (SLB). n.d. Subsea umbilical systems. Accessed August 08, 2025. <https://www.slb.com/>.
- Orcina Ltd. 2023. Waves: Wave Spectra. Accessed July 28, 2025. <https://www.orcina.com/webhelp/OrcaFlex/Content/html/Waves,Wavespectra.htm>.
- Orcina Ltd. 2023. Lines. Accessed July 28, 2025. <https://www.orcina.com/webhelp/OrcaFlex/Content/html/Lines.htm>.
- PRE Technologies. n.d. FEA based design and analysis of subsea umbilicals. Accessed August 08, 2025. <https://pretechnologies.com/>.
- TarjomeFa. n.d. Pipelines, risers, and umbilicals failures. Accessed August 08, 2025. <https://tarjomefa.com/>.
- Worzyk, Thomas. n.d. "Submarine Power Cables: Design, Installation, Repair, Environmental Aspects." Springer. Accessed August 08, 2025. <https://www.springer.com/>.
- Zhang, Y., Li, X., & Wang, J. 2021. Integrated optimisation of dynamic umbilicals. Accessed August 08, 2025. <https://www.researchgate.net/>.